Additional file 1

What Does EEG-to-fMRI Synthesis Add to EEG-Based Diagnosis? A Controlled Benchmark in Psychosis, Schizophrenia and Parkinson's Disease

Contrastive sinusoid separation: periodicity, separation property and multi-class extension

S1. Periodicity of the cosine projection

The CNN-TAG synthesizer maps its encoder output $\vec{z}$ through random Fourier features, $\cos(\omega \cdot \vec{z} + \beta)$. In order to separate data along the projection of cosines, the method has to operate in a sub-domain interval where the cosine is not periodic. Layer normalization, with centre $\frac{\pi}{2}$ and standard deviation $\frac{\pi}{2}$, maps most of the data to such an interval of the cosine (Fig. S1). This alone does not separate the classes; the contrastive loss of the main text (Methods, contrastive sinusoid separation) moves pairs of different classes to a margin m apart in this interval.

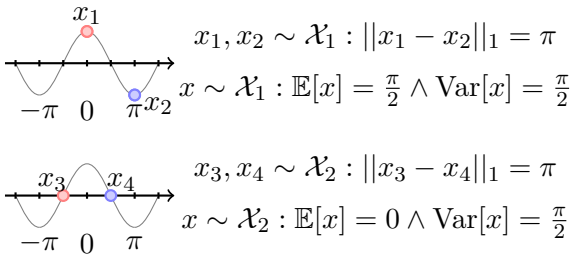


Figure S1: Description of how two similarly distributed samples, taken from $\mathcal{X}_1$ and $\mathcal{X}_2$, can lead to different portions of the cosine function image, since a sinusoid is periodic. Two distributions $\mathcal{X}_1$ and $\mathcal{X}_2$, may be mapped to the same image (second/bottom example). This is why a cosine is a shift invariant function. However, there are intervals a shift can be made and it is not invariant. Such intervals take the form $\forall i \in \mathbb{Z} : [i\pi, (i+1)\pi]$.

S2. Separation under the random cosine projection

Proposition S1 (Separation under cosine projection)**.** *Let $z_1, z_2 \in \mathbb{R}^d$ and let*

$$\phi(z) = \left[\cos(\omega_1^\top z + \beta_1), \ldots, \cos(\omega_d^\top z + \beta_d)\right]$$

with independent $\omega_i \sim \mathcal{N}(0, I_d)$ and $\beta_i \sim \mathrm{Uniform}[0, 2\pi]$. Then

$$\mathbb{E}_{\omega,\beta}\left[\phi(z_1)^\top \phi(z_2)\right] = \frac{d}{2} e^{-\frac{1}{2}\|z_1 - z_2\|_2^2},$$

so the expected similarity of the projected points decreases exponentially with their latent distance, and a contrastive margin that increases the latent distance between classes decreases their expected similarity after projection.

Proof of Proposition S1. Let

$$\phi(z) = \left[\cos(\omega_1^\top z + \beta_1), \ldots, \cos(\omega_d^\top z + \beta_d)\right],$$

where each $\omega_i \sim \mathcal{N}(0, I_d)$ and $\beta_i \sim \text{Uniform}[0, 2\pi]$. The inner product between projected vectors is:

$$\phi(z_1)^\top \phi(z_2) = \sum_{i=1}^{d} \cos(\omega_i^\top z_1 + \beta_i) \cos(\omega_i^\top z_2 + \beta_i).$$

Using the trigonometric identity:

$$\cos(a)\cos(b) = \frac{1}{2}[\cos(a-b) + \cos(a+b)],$$

we write:

$$\cos(\omega_i^\top z_1 + \beta_i)\cos(\omega_i^\top z_2 + \beta_i) = \\ \frac{1}{2}\left[\cos(\omega_i^\top (z_1 - z_2)) + \cos(\omega_i^\top (z_1 + z_2) + 2\beta_i)\right].$$

Taking the expectation over $\beta_i \sim \text{Uniform}[0, 2\pi]$, the second term vanishes:

$$\mathbb{E}_{\beta_i}[\cos(\omega_i^\top (z_1 + z_2) + 2\beta_i)] = 0.$$

Thus,

$$\mathbb{E}_{\omega,\beta}[\phi(z_1)^\top \phi(z_2)] = \frac{1}{2}\sum_{i=1}^{d} \mathbb{E}_{\omega_i}[\cos(\omega_i^\top (z_1 - z_2))].$$

Let $\delta = z_1 - z_2$. Since $\omega_i^\top \delta \sim \mathcal{N}(0, \|\delta\|_2^2)$, we use the identity:

$$\mathbb{E}_{x\sim\mathcal{N}(0,\sigma^2)}[\cos(x)] = e^{-\frac{1}{2}\sigma^2},$$

to obtain:

$$\mathbb{E}_{\omega_i}[\cos(\omega_i^\top \delta)] = e^{-\frac{1}{2}\|\delta\|_2^2}.$$

Putting it together:

$$\mathbb{E}_{\omega,\beta}[\phi(z_1)^\top \phi(z_2)] = \frac{d}{2} e^{-\frac{1}{2}\|z_1 - z_2\|_2^2}.$$

which is at most $d/2$ and decreases strictly and exponentially with $\|z_1 - z_2\|_2$, proving the result. □

S3. Extension to more than two classes

A multi-class solution is to consider the unitary circle in Fig. S2. Let the pairs of data be $(x_1, x_2, y_1, y_2, y_p)$. By definition $x_1 := \mathbf{x}_2 \in \mathbb{R}^E$. And consider the targets $y_1, y_2 \in [0, 1]^C$. Now y_p instead of being in $[0, 1]$, consider $y_p = ||argmax(y_1) - argmax(y_2)||_1^1 \times \frac{\pi}{C}$ so that the possible values are $\in \{0, \frac{\pi}{C}, \ldots, \frac{C\pi}{C}\}$. The contrastive loss now only needs one of its terms and can be written as

$$\mathcal{L}_D(x_1, x_2, y_p) = ||D(x_1, x_2) - y_p||_1^1. \tag{S1}$$

The other change that is needed is to apply the softmax at the output of the classifier, so that $y \in \Delta^C$.

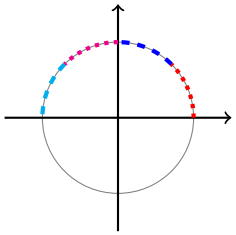


Figure S2: Hypothesis to separate multiple groups of data using the sinusoid projection. The example considers $C = 4$.